卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0316 30-B003701S-T0316

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

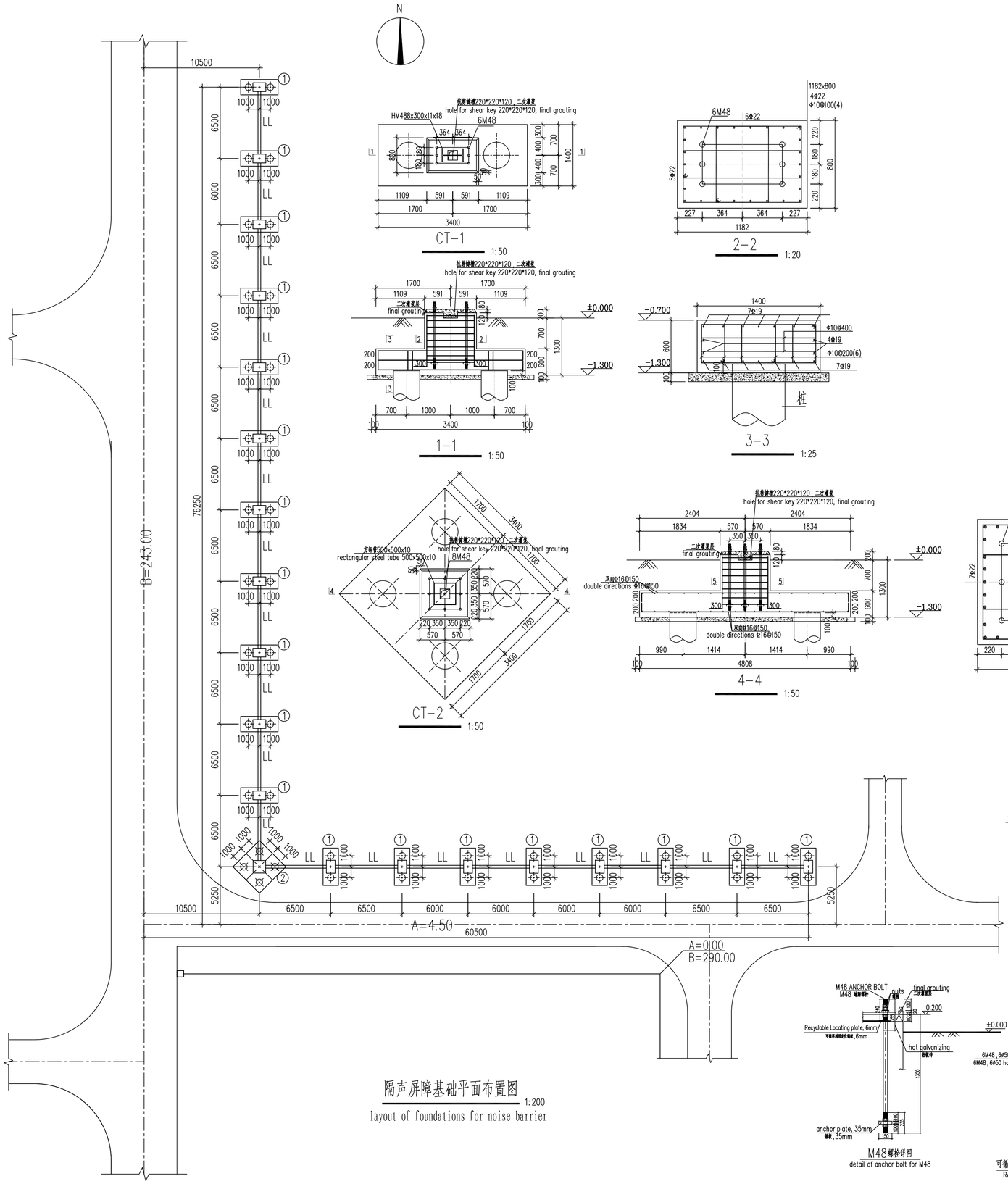
结构 专业 第 03 卷 第 16 册
Structure SPECIALITY VOLUME BOOK

卷册名称 交流滤波器场隔声屏障基础图
Volume name foundations for noise barrier in AC filter field

批 准 APPROVED BY	
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	谢财齐
图 纸 1 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-T0316-001	隔声屏障基础布置图及详图 layout and details of foundations for noise barrier	A	1		

注:



说明：

1、交流滤波器场地±0.000相当于绝对标高193.40(EGM2008高程系统)。如无特别说明，本工程标高均采用相对标高。

2、本工程基础设计依据《巴基斯坦默蒂亚里±660KV直流输电工程拉合尔±660KV换流站施工图设计阶段岩土工程勘察报告（详勘）》(B003701S-G01-01)进行。

3、本工程基础采用柱下独立承台+灌注桩的形式，桩为Φ600的机械成孔灌注桩。

4.材料

1)混凝土

基础强度等级：fc'=28MPa

二次灌浆层强度等级：fc'=35MPa

垫层强度等级：fc'=21MPa

2)钢筋：Φ-ASTM A615 Grade 40,Fy=280MPa

Φ-ASTM A615 Grade 60,Fy=420MPa

3)螺母，垫片，锚板，定位钢板：Q355B，Fy=355MPa,Fu=470MPa

锚栓：45#，Fy=355MPa,Fu=600MPa

锚栓、螺母，垫片，锚板均需热浸锌防腐。

5. 混凝土要求：

1)水胶比不大于0.50；

2)最大氯离子含量（水泥用量的百分比）：0.10；测试方法参《ASTM C 1218M》。

3)水泥型号：采用抗硫酸盐的硅酸盐水泥，水泥型号需满足《ACI 318-14》表19.3.2.1对应于暴露等级S1的水泥特类要求（ASTM C150 或ASTM 595 或ASTM C1157对应的类别）；

4)水泥中掺入矿物掺合料（矿物掺合料的掺量、使用方法和耐腐蚀性能按相应产品的使用说明并经验证后确定）。

6.钢筋保护层：承台底板底部为100mm，其余为50mm。

柱墩和连系梁为50mm.

7. 桩与承台的连接构造参见图集（10SG813）34页 桩顶与承台连接构造，桩内主筋伸入承台长度为40d。

8. 外露基础采用倒圆角处理,R=30mm

9.本工程初步设计尚未最终审定，最终方案有存在调整可能，是否按图施工由总包CET决定，由此造成的变更设计不予负责。

10.图纸上的尺寸以标注为准，不能按比例量取。

11.地脚螺栓强度等级为45#钢，需镀锌处理,镀锌范围见详图。

12.露出地面基础与地坪相接处应设置20mm胀缝，沥青混合料填缝。

13.连梁每跨均布三根Φ25 PVC管，管道联通交流滤波器场区域和站内道路侧，用于排水，管口位于场地碎石层下，需用透水材料包裹，以免异物进入。

Specification :

1. The AC filter field site level of ±0.000 is equal to the absolute elevation of 193.40 (geodestic datum system EGM2008) If not specified,the elevations of the project are adopt to the relative elevation.

2 .The foundation design of the project proceeds according to《The geotechnical engineering investigation report (Detail survey) of drawing construction design phase of ±660KV converter station in Lahore in Pakistan Matiari ±660KV HVDC project》(B003701S-G01-01).

3、The foundation of this project adopts the form of independent cap and filling pile under column. The pile is a bored cast-in-place pile with a diameter of 600.

4.MATERIALS:

1) CONCRETE:

FOR FOUNDATIONS : SPECIFIED COMPRESSIVE CONCRETE STRENGTH f'c=28MPa

FOR GROUTING : SPECIFIED COMPRESSIVE CONCRETE STRENGTH f'c=35MPa

FOR concrete cushion : SPECIFIED COMPRESSIVE CONCRETE STRENGTH f'c=21MPa

2) STEEL REINFORCING :

Φ: ASTM A615 GRADE40 , MINIMUM YIELD STRENTH Fy=280MPa

Φ: ASTM A615 GRADE60 , MINIMUM YIELD STRENTH Fy=420MPa

3) NUTS AND WASHERS, anchor plate,Locating plate :

Q355B, MINIMUM YIELD STRENTH Fy=355MPa, MINIMUM TENSILE STRENTH Fu=470MPa .

ANCHOR BOLTS: 45#, MINIMUM YIELD STRENTH Fy=355MPa, MINIMUM TENSILE STRENTH Fu=600MPa .

THE ANCHOR BOLTS, NUTS , WASHERS, anchor plate MUST BE HOT DIP GALVANIZED.

5. Requirements for Concrete:

1) Maximum water-cement ratio is 0.50;

2) Maximum water-soluble chloride ion content is 0.1; Test method refers to "ASTM C 1218M".

3) Type of Concrete: Concrete should be made of sulfate-resistance silica cement. Cement type should fulfill the requirements for concrete used in exposure grade S1, according to《ACI 318-14》table 19.3.2.1.(ASTM C150,ASTM 595 or ASTM C1157 corresponding cement type).

4) Add mineral admixtures (the dosage, using method and anti-corrosion ability should be tested according to the product manual)

6. The minimum concrete cover:

100mm for the bottom of pile cap,50mm for the rest.

50mm for pedestal and the Coupling beam.

7. The connection structure between the pile and the cap refers to the 34 page of atlas (10SG813).The main reinforcement extends into the pile cap with a length of 40d.

8. The surface of pedestal above site level should reverse corner , R=30mm.

9.As the preliminary design documents have not yet been approved finally, the project solution has probability to be modified. The design contractor has no responsibility for the project solution changed, if the construction contractor does not get authorized by CET before construction.

10.Only written dimensions shall be refered to, and don't measure in scale.

11.The anchor bolts, with 45# grade, should be galvanized according to the detail in this drawing.

12.The 20mm expansion joint should be set between the foundation and the ground, and the asphalt mixture should be filled in.

13.Three Φ25 PVC pipes should be arranged even in each span in order to connect the Ac filtering field and the road side for drainage. The mouth of the pipe should be closed by pervious material and be below the level of the gravel in the field.

隔声屏障基础工总表

SUMMARY TABLE OF FOUNDATIONS FOR NOISE BARRIER

序号 NO.	基础 FOUNDATION	基础数量 NUMBER OF FOUNDATION	基础尺寸(mm)(长*宽*高) FOUNDATION DIMENSION(mm) (LENGTH*WIDTH*DEPTH) 【图号】	单个基础混凝土体积(m^3) concrete volume of single foundation(m^3)	总混凝土量(m^3) total concrete volume (M^3)	备注 NOTE
①	CT-1	19	3.40X1.40X1.30	4.30	81.7	
②	CT-2	1	3.40X3.40X1.30	9.40	9.4	
③	LL	19			17.6	
总计：混合混凝土76m^3. TOTAL: The volume of foundations is 108.7m^3.						

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC EQUIPMENT TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 3 1 6 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	徐刚平	2019.04.30
校核 Checked	毛惠平	2019.04.30
审核 Reviewed	马建强	2019.04.30
批准 Approved	韩福彦	2019.04.30
比例 Scale	1:200	

隔声屏障基础布置图及详图

layout and details of foundations for noise barrier

图号 DWG NO. 30-B003701S-T0316-001 版本 REV. A

电气	郑煜	2019.04.30
专业 Speciality	会签 Sign	日期 Date

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